

Richness plots

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```
# libraries ----
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2     4.0.2      v tibble     3.3.1
## v lubridate   1.9.5      v tidyr      1.3.2
## v purrr       1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(readxl)

# data ----

data <- map(1:5, \(x) {
  #read all three taxa
  read_excel("data/Data2024.xlsx", sheet = x) |>
  #pivot
  pivot_longer(-1, names_to = "species") |>
  #rename first column
  rename(plot = 1)
})

#first three sheets are vascular plants, join them together
data <- bind_rows(data[1:3]) |>
  list() |>
  c(data[4:5])

data <- data |>
  #bind list to single df
  bind_rows(.id = "taxon") |>
  #rename IDs
  mutate(taxon = fct_recode(taxon,
    "Vascular plants" = "1",
    "Saproxyllic fungi" = "2",
    "Epiphytic lichens" = "3") |>
```

```

    fct_relevel("Saproxylic fungi", after = 2),
    #extract genus
    genus = word(species, 1),
    #extract epithets
    epit = word(species, 2),
    #extract site
    site = substr(plot, 1, 2) |>
      fct_recode("Prati di Tivo" = "PR",
                 "Incodaro" = "IN",
                 "Venaquaro" = "VE")
  ) |>
  arrange(taxon, species) |>
  #these were wide matrices, keep only observed entries
  filter(value > 0)

#distinct species names
distinct(data, species) |> pull() |> sort()

```

```

## [1] "Abies alba" "Acer campestre"
## [3] "Acer opalus" "Acer platanoides"
## [5] "Acer pseudoplatanus" "Actaea spicata"
## [7] "Adenostyles australis" "Adoxa moschatellina"
## [9] "Aegopodium podagraria" "Agrimonia eupatoria"
## [11] "Agrostis capillaris" "Ajuga reptans"
## [13] "Allium ursinum" "Anaptychia ciliaris"
## [15] "Anaptychia crinalis" "Aquilegia dumeticola"
## [17] "Arabis alpina" "Arabis collina"
## [19] "Arabis hirsuta" "Arctium lappa"
## [21] "Arctium minus" "Aremonia agrimonoides"
## [23] "Armillaria sp." "Arthonia didyma"
## [25] "Arthonia radiata" "Arum maculatum"
## [27] "Asperula laevigata" "Asplenium ruta-muraria"
## [29] "Asplenium trichomanes" "Astragalus glycyphyllos"
## [31] "Athallia cerinelloides" "Athallia pyracea"
## [33] "Atropa bella-donna" "Avenella flexuosa"
## [35] "Bertia moriformis" "Biscogniauxia nummularia"
## [37] "Bjerkandera adusta" "Brachypodium rupestre"
## [39] "Brachypodium sylvaticum" "Bromopsis ramosa"
## [41] "Buellia griseovirens" "Bupleurum falcatum"
## [43] "Caloplaca cerina" "Caloplaca sp."
## [45] "Calycina citrina" "Campanula glomerata"
## [47] "Campanula persicifolia" "Campanula rapunculus"
## [49] "Campanula tanfanii" "Campanula trachelium"
## [51] "Candelariella faginea" "Cardamine bulbifera"
## [53] "Cardamine enneaphyllos" "Cardamine graeca"
## [55] "Cardamine heptaphylla" "Cardamine kitaibelii"
## [57] "Carduus sp." "Carex digitata"
## [59] "Carex divulsa" "Carex flacca"
## [61] "Carex pilosa" "Carex remota"
## [63] "Carex sylvatica" "Centaurium erythraea"
## [65] "Cephalanthera damasonium" "Cephalanthera longifolia"
## [67] "Cephalanthera rubra" "Cerioporus varius"
## [69] "Chlorociboria aeruginascens" "Chondrostereum purpureum"

```

## [71]	"Circaea lutetiana"	"Cirsium arvense"
## [73]	"Clematis vitalba"	"Clinopodium menthifolium"
## [75]	"Clinopodium nepeta"	"Clinopodium vulgare"
## [77]	"Coniophora sp."	"Crataegus monogyna"
## [79]	"Cruciata glabra"	"Cruciata laevipes"
## [81]	"Crustoderma sp."	"Cyclamen repandum"
## [83]	"Cymbalaria muralis"	"Cynosurus cristatus"
## [85]	"Cystopteris fragilis"	"Cytisophyllum sessilifolium"
## [87]	"Cytisus scoparius"	"Cytisus villosus"
## [89]	"Dactylis glomerata"	"Dactylorhiza maculata"
## [91]	"Daphne laureola"	"Daphne mezereum"
## [93]	"Dianthus sylvestris"	"Diatrype disciformis"
## [95]	"Diatrype stigma"	"Digitalis micrantha"
## [97]	"Doronicum columnae"	"Drymochloa sylvatica"
## [99]	"Dryopteris filix-mas"	"Epilobium montanum"
## [101]	"Epipactis helleborine"	"Epipactis microphylla"
## [103]	"Epitrachys italica"	"Euonymus europaeus"
## [105]	"Euphorbia amygdaloides"	"Euphorbia cyparissias"
## [107]	"Euphrasia stricta"	"Fagus sylvatica"
## [109]	"Festuca heterophylla"	"Fomes fomentarius"
## [111]	"Fomitopsis pinicola"	"Fragaria vesca"
## [113]	"Fraxinus excelsior"	"Galerina marginata"
## [115]	"Galium aparine"	"Galium lucidum"
## [117]	"Galium odoratum"	"Galium rotundifolium"
## [119]	"Ganoderma applanatum"	"Geranium nodosum"
## [121]	"Geranium purpureum"	"Geranium robertianum"
## [123]	"Glaucumaria carpinea"	"Grafia golaka"
## [125]	"Gyalecta sp."	"Gyalolechia flavorubescens"
## [127]	"Hedera helix"	"Helianthemum nummularium"
## [129]	"Helleborus viridis"	"Helminthotheca echioides"
## [131]	"Hepatica nobilis"	"Heterodermia speciosa"
## [133]	"Hieracium murorum"	"Hordelymus europaeus"
## [135]	"Hypericum montanum"	"Hypericum perforatum"
## [137]	"Hypoxylon fragiforme"	"Hypoxylon sp."
## [139]	"Ilex aquifolium"	"Juniperus communis"
## [141]	"Laburnum anagyroides"	"Lachnum relicinum"
## [143]	"Lapsana communis"	"Laserpitium latifolium"
## [145]	"Lathyrus pratensis"	"Lathyrus vernus"
## [147]	"Lecania naegelii"	"Lecanora albella"
## [149]	"Lecanora argentata"	"Lecanora chlorotera"
## [151]	"Lecanora intumescens"	"Lecidella elaeochroma"
## [153]	"Lenzites betulinus"	"Lepra amara"
## [155]	"Lepraria rigidula"	"Lepraria sp."
## [157]	"Lilium bulbiferum"	"Lilium martagon"
## [159]	"Limodorum abortivum"	"Linum viscosum"
## [161]	"Lonicera alpigena"	"Lonicera etrusca"
## [163]	"Lotus corniculatus"	"Luzula forsteri"
## [165]	"Luzula multiflora"	"Luzula sylvatica"
## [167]	"Medicago lupulina"	"Megacollybia platyphylla"
## [169]	"Melanelixia glabrata"	"Melica uniflora"
## [171]	"Milium effusum"	"Moehringia trinervia"
## [173]	"Monotropa hypophaea"	"Mycelis muralis"
## [175]	"Mycena renati"	"Nemania serpens"
## [177]	"Neottia nidus-avis"	"Ochrolechia pallescens"

## [179]	"Opegrapha sp."	"Orthilia secunda"
## [181]	"Ostrya carpinifolia"	"Oxalis acetosella"
## [183]	"Parmelia sulcata"	"Parmelina spp."
## [185]	"Pertusaria coccodes"	"Pertusaria flavida"
## [187]	"Pertusaria leioplaca"	"Pertusaria pertusa"
## [189]	"Petrosedum rupestre"	"Phaeophyscia insignis"
## [191]	"Phlyctis agelaea"	"Phlyctis argena"
## [193]	"Physcia adscendens"	"Physcia aipolia"
## [195]	"Physcia biziana"	"Physcia leptalea"
## [197]	"Physcia sp."	"Physcia stellaris"
## [199]	"Physconia distorta"	"Picris hieracioides"
## [201]	"Platanthera bifolia"	"Pleurosticta acetabulum"
## [203]	"Pleurotus cornucopiae"	"Pluteus cervinus"
## [205]	"Poa compressa"	"Poa nemoralis"
## [207]	"Poa trivialis"	"Polygonatum odoratum"
## [209]	"Polyozosia hagenii"	"Polyporus tuberaster"
## [211]	"Polystichum aculeatum"	"Polystichum setiferum"
## [213]	"Potentilla micrantha"	"Prenanthes purpurea"
## [215]	"Primula vulgaris"	"Prunella vulgaris"
## [217]	"Prunus avium"	"Prunus sp."
## [219]	"Pteridium aquilinum"	"Ptilostemon strictus"
## [221]	"Pulmonaria hirta"	"Ramalina farinacea"
## [223]	"Ramalina fraxinea"	"Ranunculus lanuginosus"
## [225]	"Rhamnus alpina"	"Rosa arvensis"
## [227]	"Rosa pendulina"	"Rubus hirtus"
## [229]	"Rubus idaeus"	"Rumex acetosa"
## [231]	"Ruscus hypoglossum"	"Salix caprea"
## [233]	"Sambucus ebulus"	"Sanicula europaea"
## [235]	"Saxifraga paniculata"	"Saxifraga rotundifolia"
## [237]	"Schizopora paradoxa"	"Scleroderma verrucosum"
## [239]	"Scoliciosporum chlorococcum"	"Scutellinia sp."
## [241]	"Sedum cepaea"	"Senecio nemorensis"
## [243]	"Sesleria autumnalis"	"Sesleria nitida"
## [245]	"Silene italica"	"Solidago virgaurea"
## [247]	"Sorbus aria"	"Sorbus aucuparia"
## [249]	"Stachys sylvatica"	"Steccherinum robustius"
## [251]	"Stereum hirsutum"	"Stereum sp."
## [253]	"Stereum subtomentosum"	"Tanacetum corymbosum"
## [255]	"Taxus baccata"	"Tephromela atra"
## [257]	"Thesium sp."	"Trametes gibbosa"
## [259]	"Trametes hirsuta"	"Trametes ochracea"
## [261]	"Trametes sp."	"Trametes versicolor"
## [263]	"Trifolium campestre"	"Trifolium ochroleucon"
## [265]	"Trifolium pratense"	"Trifolium repens"
## [267]	"Urtica dioica"	"Ustulina deusta"
## [269]	"Verbascum sp."	"Veronica chamaedrys"
## [271]	"Veronica cymbalaria"	"Veronica hederifolia"
## [273]	"Veronica montana"	"Veronica officinalis"
## [275]	"Vicia cracca"	"Vicia grandiflora"
## [277]	"Vicia lathyroides"	"Viola alba"
## [279]	"Viola reichenbachiana"	"Xanthoria parietina"
## [281]	"Xerocomus subtomentosus"	"Xylaria carpophila"
## [283]	"Xylaria hypoxylon"	"Xylaria longipes"
## [285]	"Xylaria polymorpha"	"Xylaria sp."

```
## [287] "Xylariaceae sp."
```

```
#taxon species richness
```

```
data |>
```

```
  group_by(taxon) |>
```

```
  summarise(n_distinct(species), .groups = "drop")
```

```
## # A tibble: 3 x 2
```

```
##   taxon          'n_distinct(species)'
```

```
##   <fct>                                <int>
```

```
## 1 Vascular plants                      192
```

```
## 2 Epiphytic lichens                    49
```

```
## 3 Saproxylic fungi                     46
```

```
#distinct genera
```

```
distinct(data, genus) |> pull() |> sort()
```

```
##   [1] "Abies"           "Acer"           "Actaea"         "Adenostyles"
##   [5] "Adoxa"           "Aegopodium"     "Agrimonia"      "Agrostis"
##   [9] "Ajuga"           "Allium"         "Anaptychia"     "Aquilegia"
##  [13] "Arabis"          "Arctium"        "Aremonia"       "Armillaria"
##  [17] "Arthonia"        "Arum"           "Asperula"       "Asplenium"
##  [21] "Astragalus"      "Athallia"       "Atropa"         "Avenella"
##  [25] "Bertia"          "Biscogniauxia"  "Bjerkandera"    "Brachypodium"
##  [29] "Bromopsis"       "Buellia"        "Bupleurum"      "Caloplaca"
##  [33] "Calycina"        "Campanula"      "Candelariella"  "Cardamine"
##  [37] "Carduus"         "Carex"          "Centaurium"     "Cephalanthera"
##  [41] "Cerioporus"      "Chlorociboria"  "Chondrostereum" "Circaea"
##  [45] "Cirsium"         "Clematis"       "Clinopodium"    "Coniophora"
##  [49] "Crataegus"       "Cruciata"       "Crustoderma"    "Cyclamen"
##  [53] "Cymbalaria"      "Cynosurus"      "Cystopteris"    "Cytisophyllum"
##  [57] "Cytisus"         "Dactylis"       "Dactylorhiza"   "Daphne"
##  [61] "Dianthus"        "Diatrype"       "Digitalis"      "Doronicum"
##  [65] "Drymochloa"      "Dryopteris"     "Epilobium"      "Epipactis"
##  [69] "Epitrachys"      "Euonymus"       "Euphorbia"      "Euphrasia"
##  [73] "Fagus"           "Festuca"        "Fomes"          "Fomitopsis"
##  [77] "Fragaria"        "Fraxinus"       "Galerina"       "Galium"
##  [81] "Ganoderma"       "Geranium"       "Glaucomaria"    "Grafia"
##  [85] "Gyalecta"        "Gyalolechia"    "Hedera"         "Helianthemum"
##  [89] "Helleborus"      "Helminthotheca" "Hepatica"       "Heterodermia"
##  [93] "Hieracium"       "Hordelymus"     "Hypericum"      "Hypoxylon"
##  [97] "Ilex"            "Juniperus"      "Laburnum"       "Lachnum"
## [101] "Lapsana"         "Laserpitium"    "Lathyrus"       "Lecania"
## [105] "Lecanora"        "Lecidella"      "Lenzites"       "Lepra"
## [109] "Lepraria"        "Lilium"         "Limodorum"      "Linum"
## [113] "Loniceria"       "Lotus"          "Luzula"         "Medicago"
## [117] "Megacollybia"    "Melanelixia"    "Melica"         "Milium"
## [121] "Moehringia"      "Monotropa"      "Mycelis"        "Mycena"
## [125] "Nemania"         "Neottia"        "Ochrolechia"    "Opegrapha"
## [129] "Orthilia"        "Ostrya"         "Oxalis"         "Parmelia"
## [133] "Parmelina"       "Pertusaria"     "Petrosedum"     "Phaeophyscia"
## [137] "Phlyctis"        "Physcia"        "Physconia"      "Picris"
## [141] "Platanthera"     "Pleurosticta"   "Pleurotus"      "Pluteus"
```

```
## [145] "Poa" "Polygonatum" "Polyozosia" "Polyporus"
## [149] "Polystichum" "Potentilla" "Prenanthes" "Primula"
## [153] "Prunella" "Prunus" "Pteridium" "Ptilostemon"
## [157] "Pulmonaria" "Ramalina" "Ranunculus" "Rhamnus"
## [161] "Rosa" "Rubus" "Rumex" "Ruscus"
## [165] "Salix" "Sambucus" "Sanicula" "Saxifraga"
## [169] "Schizopora" "Scleroderma" "Scoliciosporum" "Scutellinia"
## [173] "Sedum" "Senecio" "Sesleria" "Silene"
## [177] "Solidago" "Sorbus" "Stachys" "Steccherinum"
## [181] "Stereum" "Tanacetum" "Taxus" "Tephromela"
## [185] "Thesium" "Trametes" "Trifolium" "Urtica"
## [189] "Ustulina" "Verbascum" "Veronica" "Vicia"
## [193] "Viola" "Xanthoria" "Xerocomus" "Xylaria"
## [197] "Xylariaceae"
```

```
#taxon genus richness
data |>
  group_by(taxon) |>
  summarise(n_distinct(genus), .groups = "drop")
```

```
## # A tibble: 3 x 2
##   taxon          'n_distinct(genus)'
##   <fct>          <int>
## 1 Vascular plants      132
## 2 Epiphytic lichens    31
## 3 Saproxylic fungi    34
```

```
#distinct epithets
distinct(data, epit) |> pull() |> sort()
```

```
##   [1] "abortivum" "acetabulum" "acetosa" "acetosella"
##   [5] "aculeatum" "adscendens" "adusta" "aeruginascens"
##   [9] "agelaea" "agrimonoides" "aipolia" "alba"
##  [13] "albella" "alpigena" "alpina" "amara"
##  [17] "amygdaloides" "anagyroides" "aparine" "applanatum"
##  [21] "aquifolium" "aquilinum" "argena" "argentata"
##  [25] "aria" "arvense" "arvensis" "atra"
##  [29] "aucuparia" "australis" "autumnalis" "avium"
##  [33] "baccata" "bella-donna" "betulinus" "bifolia"
##  [37] "biziana" "bulbifera" "bulbiferum" "campestre"
##  [41] "capillaris" "caprea" "carpineae" "carpinifolia"
##  [45] "carpophila" "cepaea" "cerina" "cerinelloides"
##  [49] "cervinus" "chamaedrys" "chlarotera" "chlorococcum"
##  [53] "ciliaris" "citrina" "coccodes" "collina"
##  [57] "columnae" "communis" "compressa" "corniculatus"
##  [61] "cornucopiae" "corymbosum" "cracca" "crinalis"
##  [65] "cristatus" "cymbalaria" "cyparissias" "damasonium"
##  [69] "deusta" "didyma" "digitata" "dioica"
##  [73] "disciformis" "distorta" "divulsa" "dumeticola"
##  [77] "ebulus" "echioides" "effusum" "elaeochroma"
##  [81] "enneaphyllos" "erythraea" "etrusca" "eupatoria"
##  [85] "europaea" "europaeus" "excelsior" "faginea"
##  [89] "falcatum" "farinacea" "filix-mas" "flacca"
```

## [93]	"flavida"	"flavorubescens"	"flexuosa"	"fomentarius"
## [97]	"forsteri"	"fragiforme"	"fragilis"	"fraxinea"
## [101]	"gibbosa"	"glabra"	"glabratula"	"glomerata"
## [105]	"glycyphyllos"	"golaka"	"graeca"	"grandiflora"
## [109]	"griseovirens"	"hagenii"	"hederifolia"	"helix"
## [113]	"helleborine"	"heptaphylla"	"heterophylla"	"hieracioides"
## [117]	"hirsuta"	"hirsutum"	"hirta"	"hirtus"
## [121]	"hypoglossum"	"hypophegea"	"hypoxylon"	"idaeus"
## [125]	"insignis"	"intumescens"	"italica"	"kitaibelii"
## [129]	"laevigata"	"laevipes"	"lanuginosus"	"lappa"
## [133]	"lathyroides"	"latifolium"	"laureola"	"leioplaca"
## [137]	"leptalea"	"longifolia"	"longipes"	"lucidum"
## [141]	"lupulina"	"lutetiana"	"maculata"	"maculatum"
## [145]	"marginata"	"martagon"	"menthifolium"	"mezereum"
## [149]	"micrantha"	"microphylla"	"minus"	"monogyna"
## [153]	"montana"	"montanum"	"moriformis"	"moschatellina"
## [157]	"multiflora"	"muralis"	"murorum"	"naegelii"
## [161]	"nemoralis"	"nemorensis"	"nepeta"	"nidus-avis"
## [165]	"nitida"	"nobilis"	"nodosum"	"nummularia"
## [169]	"nummularium"	"ochracea"	"ochroleucon"	"odoratum"
## [173]	"officinalis"	"opalus"	"pallescens"	"paniculata"
## [177]	"paradoxa"	"parietina"	"pendulina"	"perforatum"
## [181]	"persicifolia"	"pertusa"	"pilosa"	"pinicola"
## [185]	"platanoides"	"platyphylla"	"podagraria"	"polymorpha"
## [189]	"pratense"	"pratensis"	"pseudoplatanus"	"purpurea"
## [193]	"purpureum"	"pyracea"	"radiata"	"ramosa"
## [197]	"rapunculus"	"reichenbachiana"	"relicinum"	"remota"
## [201]	"renati"	"repandum"	"repens"	"reptans"
## [205]	"rigidula"	"robertianum"	"robustus"	"rotundifolia"
## [209]	"rotundifolium"	"rubra"	"rupestre"	"ruta-muraria"
## [213]	"scoparius"	"secunda"	"serpens"	"sessilifolium"
## [217]	"setiferum"	"sp."	"speciosa"	"spicata"
## [221]	"spp."	"stellaris"	"stigma"	"stricta"
## [225]	"strictus"	"subtomentosum"	"subtomentosus"	"sulcata"
## [229]	"sylvatica"	"sylvaticum"	"sylvestris"	"tanfanii"
## [233]	"trachelium"	"trichomanes"	"trinervia"	"trivialis"
## [237]	"tuberaster"	"uniflora"	"ursinum"	"varius"
## [241]	"vernus"	"verrucosum"	"versicolor"	"vesca"
## [245]	"villosus"	"virgaurea"	"viridis"	"viscosum"
## [249]	"vitalba"	"vulgare"	"vulgaris"	

```
#check incerta
```

```
incerta_index <- c("", "sp.", "spp.")
```

```
incerta <- data |>
```

```
  filter(epit %in% incerta_index) |>
```

```
  distinct(species, genus, taxon)
```

```
incerta
```

```
## # A tibble: 19 x 3
```

##	species	genus	taxon
##	<chr>	<chr>	<fct>
##	1 Carduus sp.	Carduus	Vascular plants

```
## 2 Prunus sp.      Prunus      Vascular plants
## 3 Thesium sp.     Thesium     Vascular plants
## 4 Verbascum sp.   Verbascum   Vascular plants
## 5 Caloplaca sp.   Caloplaca   Epiphytic lichens
## 6 Gyalecta sp.    Gyalecta    Epiphytic lichens
## 7 Lepraria sp.    Lepraria    Epiphytic lichens
## 8 Opegrapha sp.   Opegrapha   Epiphytic lichens
## 9 Parmelina spp.  Parmelina   Epiphytic lichens
## 10 Physcia sp.     Physcia     Epiphytic lichens
## 11 Armillaria sp.  Armillaria  Saproxylic fungi
## 12 Coniophora sp. Coniophora  Saproxylic fungi
## 13 Crustoderma sp. Crustoderma Saproxylic fungi
## 14 Hypoxylon sp.  Hypoxylon   Saproxylic fungi
## 15 Scutellinia sp. Scutellinia Saproxylic fungi
## 16 Stereum sp.     Stereum     Saproxylic fungi
## 17 Trametes sp.    Trametes    Saproxylic fungi
## 18 Xylaria sp.     Xylaria     Saproxylic fungi
## 19 Xylariaceae sp. Xylariaceae Saproxylic fungi
```

#check congenetics of incerta

```
data |>
  filter(genus %in% incerta$genus) |>
  distinct(species, taxon) |>
  print(n = 50)
```

```
## # A tibble: 38 x 2
##   species      taxon
##   <chr>      <fct>
## 1 Carduus sp.  Vascular plants
## 2 Prunus avium Vascular plants
## 3 Prunus sp.   Vascular plants
## 4 Thesium sp.  Vascular plants
## 5 Verbascum sp. Vascular plants
## 6 Caloplaca cerina Epiphytic lichens
## 7 Caloplaca sp. Epiphytic lichens
## 8 Gyalecta sp.  Epiphytic lichens
## 9 Lepraria rigidula Epiphytic lichens
## 10 Lepraria sp.  Epiphytic lichens
## 11 Opegrapha sp. Epiphytic lichens
## 12 Parmelina spp. Epiphytic lichens
## 13 Physcia adscendens Epiphytic lichens
## 14 Physcia aipolia Epiphytic lichens
## 15 Physcia biziana Epiphytic lichens
## 16 Physcia leptalea Epiphytic lichens
## 17 Physcia sp.   Epiphytic lichens
## 18 Physcia stellaris Epiphytic lichens
## 19 Armillaria sp. Saproxylic fungi
## 20 Coniophora sp. Saproxylic fungi
## 21 Crustoderma sp. Saproxylic fungi
## 22 Hypoxylon fragiforme Saproxylic fungi
## 23 Hypoxylon sp.  Saproxylic fungi
## 24 Scutellinia sp. Saproxylic fungi
## 25 Stereum hirsutum Saproxylic fungi
```



```
## 26 Stereum sp.          Saproxylic fungi
## 27 Stereum subtomentosum Saproxylic fungi
## 28 Trametes gibbosa     Saproxylic fungi
## 29 Trametes hirsuta     Saproxylic fungi
## 30 Trametes ochracea    Saproxylic fungi
## 31 Trametes sp.        Saproxylic fungi
## 32 Trametes versicolor  Saproxylic fungi
## 33 Xylaria carpophila   Saproxylic fungi
## 34 Xylaria hypoxylon    Saproxylic fungi
## 35 Xylaria longipes     Saproxylic fungi
## 36 Xylaria polymorpha   Saproxylic fungi
## 37 Xylaria sp.         Saproxylic fungi
## 38 Xylariaceae sp.      Saproxylic fungi
```

#those that have no congeners

```
keep_genus <- data |>
  filter(genus %in% incerta$genus) |>
  distinct(genus, species, taxon) |>
  add_count(genus) |>
  filter(n == 1)
```

keep_genus

```
## # A tibble: 11 x 4
##   genus      species      taxon      n
##   <chr>      <chr>      <fct>    <int>
## 1 Carduus    Carduus sp.  Vascular plants 1
## 2 Thesium    Thesium sp.  Vascular plants 1
## 3 Verbascum  Verbascum sp. Vascular plants 1
## 4 Gyalecta   Gyalecta sp. Epiphytic lichens 1
## 5 Opegrapha  Opegrapha sp. Epiphytic lichens 1
## 6 Parmelina  Parmelina spp. Epiphytic lichens 1
## 7 Armillaria Armillaria sp. Saproxylic fungi 1
## 8 Coniophora Coniophora sp. Saproxylic fungi 1
## 9 Crustoderma Crustoderma sp. Saproxylic fungi 1
## 10 Scutellinia Scutellinia sp. Saproxylic fungi 1
## 11 Xylariaceae Xylariaceae sp. Saproxylic fungi 1
```

These should be kept because they are the only ones in their genus. Now, let's check the ones with more entries:

```
data |>
  filter(genus %in% incerta$genus) |>
  distinct(genus, species, taxon) |>
  add_count(genus) |>
  filter(n > 1) |>
  print(n = 50)
```

```
## # A tibble: 27 x 4
##   genus      species      taxon      n
##   <chr>      <chr>      <fct>    <int>
## 1 Prunus     Prunus avium  Vascular plants 2
```

##	2	Prunus	Prunus sp.	Vascular plants	2
##	3	Caloplaca	Caloplaca cerina	Epiphytic lichens	2
##	4	Caloplaca	Caloplaca sp.	Epiphytic lichens	2
##	5	Lepraria	Lepraria rigidula	Epiphytic lichens	2
##	6	Lepraria	Lepraria sp.	Epiphytic lichens	2
##	7	Physcia	Physcia adscendens	Epiphytic lichens	6
##	8	Physcia	Physcia aipolia	Epiphytic lichens	6
##	9	Physcia	Physcia biziana	Epiphytic lichens	6
##	10	Physcia	Physcia leptalea	Epiphytic lichens	6
##	11	Physcia	Physcia sp.	Epiphytic lichens	6
##	12	Physcia	Physcia stellaris	Epiphytic lichens	6
##	13	Hypoxylon	Hypoxylon fragiforme	Saproxylic fungi	2
##	14	Hypoxylon	Hypoxylon sp.	Saproxylic fungi	2
##	15	Stereum	Stereum hirsutum	Saproxylic fungi	3
##	16	Stereum	Stereum sp.	Saproxylic fungi	3
##	17	Stereum	Stereum subtomentosum	Saproxylic fungi	3
##	18	Trametes	Trametes gibbosa	Saproxylic fungi	5
##	19	Trametes	Trametes hirsuta	Saproxylic fungi	5
##	20	Trametes	Trametes ochracea	Saproxylic fungi	5
##	21	Trametes	Trametes sp.	Saproxylic fungi	5
##	22	Trametes	Trametes versicolor	Saproxylic fungi	5
##	23	Xylaria	Xylaria carpophila	Saproxylic fungi	5
##	24	Xylaria	Xylaria hypoxylon	Saproxylic fungi	5
##	25	Xylaria	Xylaria longipes	Saproxylic fungi	5
##	26	Xylaria	Xylaria polymorpha	Saproxylic fungi	5
##	27	Xylaria	Xylaria sp.	Saproxylic fungi	5

Of those, *Prunus* is kept because it refers to seedlings which are distinct from those of *Prunus avium*. *Lepraria* is kept because it was different from *Lepraria rigidula*, while *Physcia* and *Caloplaca* should be removed.

```
keep_genus <- c(keep_genus$genus, "Lepraria", "Prunus")
```

```
#keep only those incerta
```

```
incerta <- incerta |>
  filter(!(genus %in% keep_genus))
```

```
data <- data |>
  anti_join(incerta)
```

```
## Joining with 'by = join_by(taxon, species, genus)'
```

```
# plot! -----
```

```
richness <- data |>
  group_by(site, taxon) |>
  summarise(rich = n_distinct(species), .groups = "drop")
```

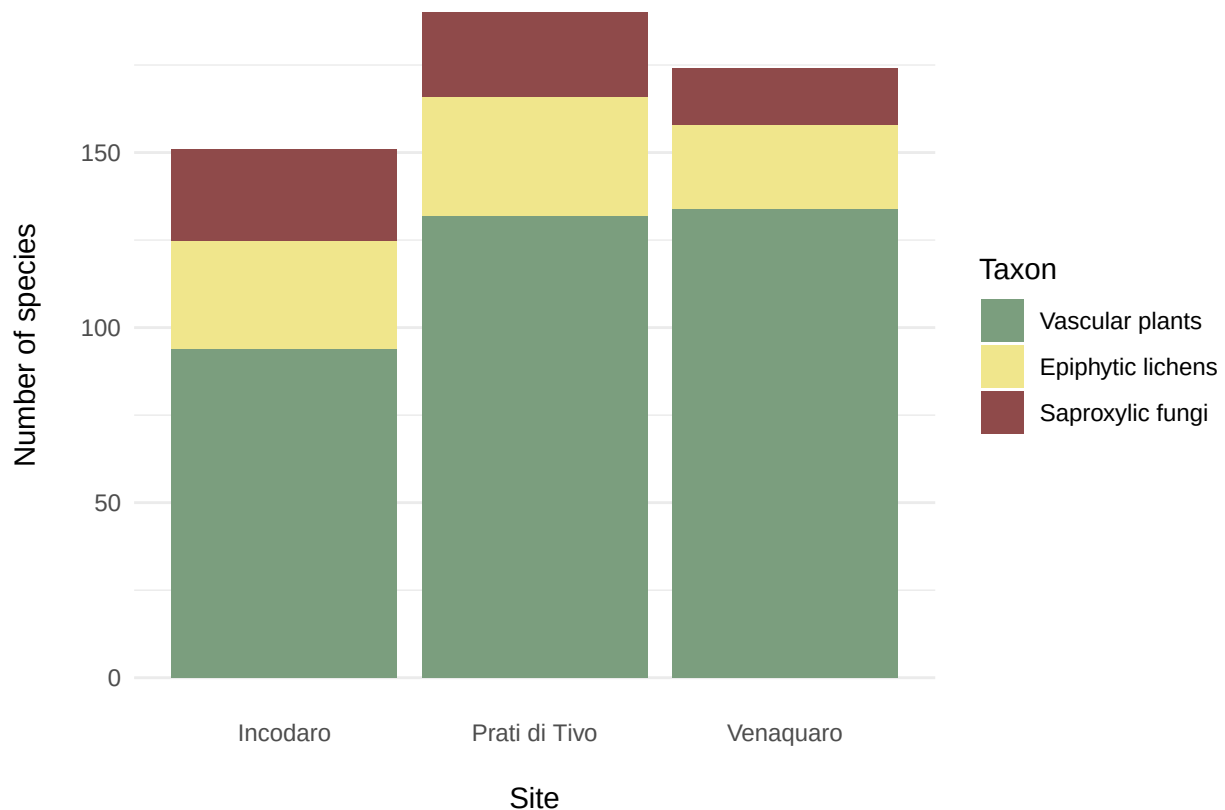
```
richness
```

```
## # A tibble: 9 x 3
```

```
##   site      taxon      rich
##   <fct>    <fct>    <int>
## 1 Incodaro Vascular plants    94
```

## 2	Incodaro	Epiphytic lichens	31
## 3	Incodaro	Saproxylic fungi	26
## 4	Prati di Tivo	Vascular plants	132
## 5	Prati di Tivo	Epiphytic lichens	34
## 6	Prati di Tivo	Saproxylic fungi	24
## 7	Venaquaro	Vascular plants	134
## 8	Venaquaro	Epiphytic lichens	24
## 9	Venaquaro	Saproxylic fungi	16

```
richness |>
  ggplot(aes(x = site, y = rich, fill = taxon)) +
  geom_col(position = position_stack(reverse = TRUE)) +
  scale_fill_discrete(palette = c("#7B9E7E", "#F0E68C", "#8F4A4A")) +
  theme_minimal() +
  labs(
    x = "Site",
    y = "Number of species",
    fill = "Taxon"
  ) +
  theme(
    axis.title.x = element_text(margin = margin(t = 15)),
    axis.title.y = element_text(margin = margin(r = 15)),
    plot.title = element_text(hjust = 0.5, size = 16, face = "bold", margin = margin(b = 15)),
    panel.grid.major.x = element_blank()
  )
```



```
ggsave("richness_stack.png", width = 119, height = 73, scale = 1.2, units = "mm", bg = "white")  
ggsave("Fig2.eps", width = 119, height = 73, scale = 1.2, units = "mm", bg = "white")
```